Credit Risk Portfolio Analytics — Architecture

1 · Raw data	2 · Ingestion	3 · Analytics & model	4 · Product
5 synthetic CSVs customers · loans · bureau · repayments · collections **Tool:** pandas	Schema checks · normalization · missingness flags one-to-many aggregation to account grain **Output:** analysis_dataset.parquet	Portfolio cuts · engineered repayment/bureau features logistic baseline + histogram gradient boosting **Tools:** pandas · scikit-learn	Streamlit portfolio/account views · Plotly charts risk score, drivers, filters and review workflows **Hosting:** Streamlit Community Cloud

Embedded AI controls

Ask the Portfolio	Question → allow-listed intent/dimension → pandas aggregation over active filters → verified fact → optional OpenAI call for wording only. The LLM receives the question and computed fact, never raw customer rows. Unsupported questions are declined; deterministic answer remains available without a key.
Anomaly detection	Account features → loan-type × income-band peer cohort → median/MAD robust score → reason code → analyst Confirm/Dismiss. No LLM and no automated adverse action. Review state should move to an authenticated database in production.

Artifacts & governance

Artifacts: clean/scored Parquet, fitted joblib pipeline, metrics JSON, data-quality report. Secrets: OPENAI_API_KEY only in the host secret store. Monitoring: data quality → drift/calibration/discrimination → review outcomes → controlled retraining and approval. Raw IDs remain in-app; the LLM receives only aggregate facts.